Rishi Kolluru

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Education

University at Buffalo, SUNY — Buffalo, NY
M.S. in Computer Science and Engineering — *Aug. 2025 – Dec. 2026 (Expected)*

Institute of Aeronautical Engineering (IARE) — Telangana, India
B.Tech. in Computer Science and Engineering — *Dec. 2021 – Jul. 2025*

Technical Skills

Programming Languages: Python, Java, SQL, C, JavaScript
ML / LLM: PyTorch, Hugging Face Transformers, vLLM, ONNX Runtime, scikit-learn, TensorFlow, MLflow, Ollama
Retrieval: FAISS, BM25, Sentence-Transformers, cross-encoders, HNSW, IVF-PQ, reciprocal rank fusion
Systems: FastAPI, asyncio, Docker, GitHub Actions, PostgreSQL, Redis, Git, Linux, REST APIs

Projects

LLM Inference Gateway — github.com/RishDevs/llm-inference-gateway
Python, PyTorch, Hugging Face, FastAPI, asyncio, Docker, vLLM

- Built a FastAPI inference gateway for Qwen2.5-1.5B with continuous micro-batching, FP16/SDPA inference, and token streaming; in closed-loop load tests at **32 concurrent users** on a T4, it delivered **12.2**× the throughput of a request-at-a-time baseline, cut queue-driven median time to first token from **109 s to 150 ms**, and reached **39%** of vLLM's throughput (3-run means).
- Implemented token-budget admission control with preemption and backpressure; under a **4**× long-context burst it held **zero OOMs** at **7.0 GB** peak memory, while the unguarded server crashed **3 times** and failed **77%** of requests.
- Profiled decode with PyTorch Profiler and added cost-model-driven length lanes, cutting padded KV tokens **62%** and improving throughput **1.13**× and p99 latency **12%** on long mixed-length prompts.

Hybrid Search & Ranking Engine — github.com/RishDevs/hybrid-search-engine
Python, FAISS, Sentence-Transformers, cross-encoders, scikit-learn, FastAPI, GitHub Actions

- Built a hybrid retriever combining from-scratch BM25, bge-small embeddings in FAISS, weighted reciprocal rank fusion, and cross-encoder reranking to handle both exact terms and paraphrases; with fusion weights and reranker tuned on dev splits only, it improved nDCG@10 by **5.5 points** over dense retrieval and **12.6** over BM25 across **3 BEIR datasets** ($p < 10^{-5}$).
- Cut mean query latency **57%** with a ridge-regression router that predicts each query's reranking gain from **8** retrieval signals and sends only **49%** of queries to the **1.6 s/query** CPU cross-encoder, recovering **69%** of the gain (vs. **49%** for random routing).
- Benchmarked **12** FAISS configurations on a **200K-document** corpus: HNSW was **51**× faster than exact search at **0.97 Recall@10**, and binary quantization with rescoring shrank the index **32**× at a **0.01-point** nDCG loss; added a GitHub Actions gate that fails PRs on retrieval drops over **0.5 points**, with reranker score caching cutting gate time from **88 min to 39 s**.

LLM-to-Small-Model Cascade — github.com/RishDevs/llm-small-model-cascade
Python, PyTorch, Hugging Face, ONNX Runtime, Ollama, FastAPI

- Built a confidence-routed cascade for CLINC150 intent classification in which an INT8 DistilBERT student handles confident queries and defers the rest to a 27.8B Qwen teacher; under a **1%** teacher-disagreement budget, it cut LLM calls **68%** at teacher-level accuracy (**89.3%** vs. **89.1%**).
- Trained the student without human annotation by distilling the teacher's zero-shot labels, generated locally through Ollama with JSON-schema constrained decoding (**0** invalid outputs, no API cost); a teacher ablation showed the 27.8B model avoided **68%** of LLM calls vs. **45%** with a 7B teacher.
- Exported the student to ONNX with dynamic INT8 quantization, shrinking it **4**× (257 → 65 MB) and serving at **1.9 ms p50** / **3.1 ms p95** on CPU, **1.9**× faster than FP32 with **98.3%** prediction agreement.

Experience

Pullulate Private Limited (College-Incubated Startup) — Feb. 2023 – Feb. 2025
Software Engineering Intern – Data & ML — *Remote / Hyderabad, India*

- Built Pandas and scikit-learn preprocessing workflows to standardize GI-tagged product metadata, detect duplicate listings, and prepare datasets, achieving **95%** product-category mapping accuracy.
- Developed TF-IDF–based product classification and content-based recommendation prototypes using Logistic Regression and cosine similarity; achieved **94%** held-out accuracy and **89% Precision@5**, respectively.
- Integrated the OpenAI API into a shopping-assistant prototype that answered product queries, explained GI certifications, and supported product discovery, with **92%** response accuracy on a manually evaluated test set.

CETA – Computer Science Club — Apr. 2023 – Oct. 2024
Analytics Lead — *Institute of Aeronautical Engineering, India*

- Analyzed **1,200+** coding-assessment records across **8** DSA topics with Python and Pandas and built a Random Forest proficiency classifier (**94%** held-out accuracy), surfacing results in **3** Power BI dashboards.